

Predicting Electric Vehicle Adoption: A Machine Learning Approach Using Socioeconomic and Infrastructure Data

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Abstract—The adoption of EVs are dependent on socioeconomic factors and existing infrastructure that is available to support it. This research employs regression machine learning techniques, to analyze the relationship between EV population within a specified county and key socioeconomic factors such as, GDP, population, population density, and EV chargers. This research utilizes datasets from multiple states in the U.S., including Washington, California, Texas, Florida, and New York. It implements the XGBoost, Random Forest, Linear Regression, and Support Vector Regression models to predict EV populations. Our findings indicate that existing EV infrastructure such as chargers heavily influence EV population rates, with XGBoost demonstrating the highest predictive accuracy with an R^2 score of 0.841 and an accuracy score of 89.58% (1000 car tolerance). The results underscore the importance of infrastructure development to enhance EV adoption.

Index Terms—Electric Vehicles, Machine Learning, Regression, Predictive Modeling

I. INTRODUCTION

The global shift towards electric vehicles has increased in multiple countries. With electric vehicle companies making high-risk expansion domestically and internationally, it raises the question of what influences EV¹ adoption in several of these markets. One probable cause is the economic health of a country, represented by GDP² and many other socioeconomic factors, such as population, and EV infrastructure. This research seeks to find a correlation of these factors with the population of EVs within a specified county in the U.S.³. However, additional factors such as culture or government policies may

also play a role in EV adoption. Despite these complexities, we will approach the data collection from various data sharing platforms to ensure the consistency and reliability of the data. By cross-referencing multiple data sources, we aim to build a robust dataset that can provide accurate insights with minimal outliers. We aim that the outcome shows a correlation between socioeconomic factors of a country and its EV population. In addition, predict future populations based on the data we analyzed.

II. RELATED WORK

As the global trend of electric vehicles increases, there are numerous reports and data analysis that suggest a link between sales in the global EV market with the GDP of countries. More specifically, sales in the EV market are dominated by countries with the highest GDP. These countries have the largest car markets, namely the United States, China, and developed countries in Europe such as Norway, Sweden, and the Netherlands. Together, they made up almost 95% of global electric vehicle market sales in 2023.

There are related works and studies that utilizes a logistic regression algorithm to determine the adoption of EVs within a specified region. Dixit and Singh [7] developed a predictive machine learning model that determined whether a person would “Buy” or “Won’t buy” an electric vehicle in India. It was found that age, gender, income, level of environmental concern, vehicle cost, and mass behaviors are the features most associated with whether or not users purchase a vehicle in India. [7]. To predict early adopters of EVs, Rubens [25] used K-means to determine that between 84% and 94% of respondents would consider an EV below €30,000. Bas and

¹Electric Vehicles

²Gross Domestic Product

³United States

others [2] utilized Support Vector Machines, Neural Networks, and XGBoost to determine the amount of EV adopters and non-adopters within the U.S.. Additionally Jia and others [13] stated that increasing gasoline prices were a contributing factor as well to the adoption of electric vehicles. Jia and other's implementation indicated that a Decision Tree model is better performing compared to Support Vector Machines, Naïve Bayes, and Logistic Regression Model.

Besides EV populations, machine learning has been used for many other applications as well. Sheng and Xiao [19] used a Support Vector Machine to predict EV battery health and charge state. While Zahid and others [24] used an Artificial Neural Network to estimate the state of the batteries. Das et al. [6] used K-nearest neighbors, Support Vector Regressor, Decision Tree, and Random Forest to analyze and determine the capacity loss of EV batteries. Similarly, to predict the battery capacity for EV vehicles, Zhao et al. [27] used machine learning techniques and Bayesian Regression to determine EV range under different driving conditions. Sun et al. [21] conducted research on predicting the driving range of EVs, by using Linear Regression, Gradient Boosting, Decision Tree, based on weather, building density, SOC, voltage, current, and temperature.

Fukushima and others [10] predicted the energy consumption for new EV models, and with this data they gave recommendation to EV users on the next charging location they should stop at. Similarly, Basso et al. [3] took into account the limited range of commercial EVs to find the best routes through traffic conditions by using a probabilistic Bayesian machine.

Yavasoglu and others [23] used an Artificial Neural Network as well to estimate the range of an EV, additionally they also estimated the road type an EV is travelling over with a decision tree. Shahriar et al. [18] predicted EV charging behaviors with random forest, SVMs, XGBoosts, and Deep Neural Networks to improve scheduling algorithms. To predict the duration of charge and energy consumption, Chung [4] used Support Vector Machines, Random Forest, and Diffusion-based kernel density estimators and trained these models on past charging and consumption records. Majidpour et al. [15] also predicted energy consumption at charging outlets by using K-nearest neighbor, and also the next-day demands of a charging station based on its previous records by using Support Vector Machines and Random Forest [14]. Zhang et al. [26] used a Neural Network to predict and forecast EV energy consumption to serve as a foundation to assist other researchers in optimizing their urban transport systems for EVs.

Xu [22] used Support Vector Machines to predict the arrival and departure time of EV commuters, similarly Frendo et al. [9] also predicted departure times of EVs with XGBoosts, Artificial Neural Networks, and Support Vector Machines as well. In order to predict the availability of an EV in providing vehicle to home services, Aguilar-Dominguez et al. [1] used optimization models such as Pyomo and Gurobi frameworks to limit the consumer's electricity bill by optimizing charging

and discharging while at the consumer's home.

The main difference between our solution and other previously conducted studies is this research will focus on a much more in-depth analysis on a country's regional economy and how these factors play a significant role in EV adoption. Additionally, we will also factor in existing EV infrastructure, as that is a factor many EV users consider before purchasing an electric vehicle in their area. Our solution aims to be more centered around the country rather than the personal factors and decision making process that many EV consumers go through or technical factors about EV range or battery state.

III. METHODOLOGY

Machine learning models such as Random Forest, Linear Regression, Gradient Boosting, and Support Vector Regression will be used for this research due to their regression capabilities. Multiple features related to the socioeconomic factors such as GDP, population, and population density will be used as well. Other important factors that must be considered would be the implementation of EV infrastructure to support EV adoption. Therefore, the amount of EV charger stations within a given county would also play a pivotal role in allowing us to predict EV adoption.

A. Dataset Overview and Description

CSV datasets from kaggle and government census websites will be used for this document. All of the datasets used within our research dated the year 2020 and above only. The electric vehicle population in Washington, U.S., is obtained from a Kaggle website that freely offers this data [12]. California and Florida electric vehicle populations are obtained from a state registration website that collects EV registration data from several states [16]. New York electric vehicle population is obtained from a DMV website by New York State [20]. And lastly, Texas electric vehicle population dataset is obtained from a Texas-Based NGO that promotes electric vehicle adaptation within Texas cities [5], additionally it is also sourced and cross checked with another government census website [16].

For the GDP feature that will be used for the prediction models, the total GDP of each county within the state of Washington, California, Florida, New York, and Texas will be obtained from local government census website [11]. Alongside population and population density information as well [17]. Datasets about the total available public electric vehicle charger stations will be obtained from a government census from the U.S. Department of Energy [8]. The following figures are all snippets of the datasets used in this research.

This research has made the correlation that EV populations tend to be located within counties that are more densely populated. Counties known for their density and high populations tend to contain higher GDPs and operate businesses in the tertiary sectors (Finances, administration, education, real-estate, technology). Counties with lower GDPs, density, and population have lower EV populations as well. This low population may be the result of the need for other modes

of transportation like diesel trucks and gasoline cars for harvesting and other primary sector related businesses such as animal farming. It can also be due to the lack of EV supporting infrastructure such as chargers.

B. Dataset Pre-processing

This research uses a dataset from 2024 for EV populations within the state of Washington. The dataset for EV populations in California has only been recently updated in 2020, Florida in 2021, New York in 2024, and Texas in 2024. All datasets have been cleaned and pre-processed by dropping any NaN and Null values as well as removing any undesirable data formats within specific columns (Removing integers in columns for strings, and vice-versa). Additional testing is conducted on the data to ensure that there are no abnormalities located within it.

Many of the used EV population datasets use cities and postcodes to determine the location of the registered vehicle, resulting in the need to also obtain datasets of cities and postcodes within these states and match them with their respective counties, allowing for the registration data to be grouped based on counties rather than the original cities or postcodes format. Lastly, all counties and their respective features are placed into a large table consisting of 479 rows, with the columns of ‘Number of electric vehicle’, ‘Total Chargers’, ‘GDP’, ‘Population’, ‘Land Area’, ‘Population Density’. This final table will later be used for the regression machine learning models.

Table I shows a snippet of the electric vehicle population datasets used for this research. Table 1 is a list of each county and their corresponding electric vehicle population within the state of Washington. This data has been aggregated and grouped together based on County, the original dataset contained individual data entries from each EV in the state of Washington.

TABLE I
ELECTRIC VEHICLE POPULATION DATASET.

County	Car Count
King	99076
Snohomish	22643
Pierce	14932
Clark	11328
Thurston	6969
Adams	55
Lincoln	51
Ferry	35
Columbia	13
Garfield	2

Table II shows a snippet of the population and population density of each county. There were no NaN values or abnormalities within this dataset, and only the population and density columns will be used for the machine learning models. Table III displays a snippet of the GDP in county within Texas 2023. Years 2019, 2021, and 2022 have been dropped from the database. This research will utilize only GDP data from 2023 and 2024 to stay recent.

TABLE II
COUNTY POPULATION AND DENSITY DATASET

pop2024	county	landArea	density
2277449	King	2116	1076.299149
930913	Pierce	1668	558.101319
849070	Snohomish	2087	406.837566
553170	Spokane	1764	313.588435
525563	Clark	628	836.883758
11863	Lincoln	2311	5.133276
7526	Ferry	2203	3.416251
4842	Wahkiakum	263	18.410646
4080	Columbia	869	4.695052
2369	Garfield	711	3.331927

TABLE III
2023 TEXAS COUNTY GDP

County	2023 GDP
Anderson	2164042
Andrews	4128580
Angelina	3512511
Aransas	786741
Archer	340087
Wood	1582826
Yoakum	1983345
Young	1152963
Zapata	586267
Zavala	804738

For the total chargers feature, the data is extracted from a government census that logged each electric vehicle charger present within each city. Table IV displays the simplification this data, the entire dataset was grouped based on counties. Different levels of charging such as EV Level-1 EVSE, EV Level-2 EVSE, and EV DC Fast, were combined into one column ‘Total Chargers’ as each level has the same purpose of charging a vehicle, however with alternative speeds.

TABLE IV
TOTAL CHARGERS WITHIN EACH CALIFORNIA COUNTY

County	Total
Alameda	6478
Alpine	12
Amador	72
Butte	259
Calaveras	41
Tulare	488
Tuolumne	100
Ventura	1531
Yolo	810
Yuba	112

Lastly, all datasets have been combined onto one singular table to be used by the machine learning models to make predictions on electric vehicle populations. Table V and VI displays the final dataset to be used. Due to the width of the original table, it has been split into two to fit.

C. Model and Techniques

Tools that are used for this research include Pandas, Matplotlib, Scikit-learn, and Numpy. Pandas is a Python library that allows us to analyze, clean, and manipulate our datasets in the manner that we require.

TABLE V
TABLE USED FOR MACHINE LEARNING MODELS (PART 1)

County	Number of Vehicles	Total Chargers	GDP
Los Angeles	581479.000000	60698.000000	801967353
Orange	263287.000000	8522.000000	273063592
Santa Clara	223206.000000	23365.000000	390234709
San Diego	175122.000000	10982.000000	261671687
Alameda	134796.000000	6478.000000	150532046
Osceola	0.000000	0.000000	14415205
Lafayette	0.000000	0.000000	210912
Hamilton	0.000000	0.000000	588042
Throckmorton	0.000000	0.000000	160434
St. Lawrence	0.000000	0.000000	4847243

TABLE VI
TABLE USED FOR MACHINE LEARNING MODELS (PART 2)

County	Population	Land Area	Population Density
Los Angeles	9606925	4060	2366
Orange	3121138	793	3935
Santa Clara	1876849	1291	1453
San Diego	3262770	4210	775
Alameda	1616117	737	2192
Osceola	453025	1328	341
Lafayette	8379	543	15
Hamilton	13748	514	26
Throckmorton	1513	913	1
St. Lawrence	106791	2680	39

With pandas, we will be able to categorize and narrow down broad and general tables of EV statistics into focused categories that satisfy our goals. Matplotlib allows us to visualize our data into clean and simple graphs for representation and easy analysis. Scikit-learn is another python machine-learning library that will help create and fit our models according to our datasets. Numpy provides access to a large collection of mathematical functions that will be useful in computing and cleaning up datasets and for more math-centered usage.

A Random Forest, Linear Regression, Gradient Boosting Machine, and Support Vector Regression models are used due to its effectiveness in predicting number values from features who have a linear relationship with the target column. In the datasets, the model will attempt to predict the total population of electric vehicles in a given county with only 4 features, the amount of EV charging stations, GDP, population, and population density. Each model will be fed and trained on the same dataset.

D. Evaluation Techniques

To evaluate the performance of each machine learning model, the MSE⁴, RMSE⁵, R^2 score, MAE⁶, P-value, Spearman Correlation, accuracy (1000 car tolerance), and feature importance were all obtained from the available model for direct analysis and comparison with one another. The final predictions were also analyzed to determine if there are any anomalies within a model's prediction. Additionally, all the predicted performance metrics will be placed into a single table for direct comparison and easier viewing.

⁴Mean Squared Error

⁵Root Mean Squared Error

⁶Mean Absolute Error

IV. RESULTS AND DISCUSSION

A. Model Performance Metrics

TABLE VII
PERFORMANCE METRICS OF MACHINE LEARNING MODELS (PART 1)

Model	MSE	RMSE	R^2 Score	MAE
XGBoost	6762762.002	2600.531	0.841	1069.648
Random Forest	11255710.83	3354.953	0.817	1110.46
Linear Regression	6.74E+08	25956.830	0.82	5544.774
Support Vector	117934217.2	10859.752	0.532	2601.23

TABLE VIII
PERFORMANCE METRICS OF MACHINE LEARNING MODELS (PART 2)

Model	Accuracy	Spearman Correlation	P-Value
XGBoost	89.58%	0.932	7.24E-22
Random Forest	84.38%	0.921	3.12E-40
Linear Regression	51.04%	0.841	7.89E-27
Support Vector	83.33%	0.886	2.93E-97

XGBoost gives relatively accurate predictions, however, a high MSE indicates a large variance between predicted and actual values. Meaning the model is not perfectly capturing the data. However, this can be overlooked due to the large scale of the data, as well as the relatively low RMSE and MAE score that XGBoost has. Additionally, it has a very strong variance with an R-squared score of 84.1%, and a high consistency and statistical significance as shown by XGBoost's very low P-value. And when given a leniency of 1000 vehicles, its accuracy score of 89.58% indicates effective prediction.

Random Forest has a larger difference between predicted and actual values as shown by its higher MSE and RMSE. However, Random Forest's MAE performance does not deteriorate as much, remaining close to XGBoost's performance. It explains its variance well and also show good consistency and statistical significance with its high R-squared score and low P-value. Lastly, Random Forest has a similar accuracy score of 84.38% to that of XGBoost.

Linear Regression has a significantly higher MAE compared to other models, indicating that it is less accurate. An MSE and RMSE of $6.74E + 08$ and 25956.830 respectively is the worse performing of all models, showing a clear reduction in performance and accuracy. Additionally, it also has the worse accuracy score with 51.04%. However, Linear Regression still continues to have a high R-squared score and low P-value that is the same with all three other models.

Support Vector Regression has the lowest R-squared score of all models, a mediocre accuracy as shown with its high MSE, RMSE, and MAE. Despite this, it still had a high spearman correlation and low P-value, indicating that there is statistically significant predictions but lacking performance.

B. Feature Importance

As seen in Table IX, GDP and Total Chargers are the most important features and show the strongest correlations with the number of electric vehicles for the XGBoost model. Indicating that the economy and availability of chargers

TABLE IX
XGBOOST FEATURE IMPORTANCE

Feature Number	Feature Name	F-Score
FO	GDP	528.0
F3	Total Chargers	326.0
F2	Population Density	257.0
F1	Population	195.0

TABLE X
RANDOM FOREST FEATURE IMPORTANCE

Feature	Importance
GDP	0.019159
Population	0.083599
Population Density	0.005412
Total Chargers	0.891831

Random Forest continues the trend with the total chargers feature being the most significant predictor, which implies an increase in chargers would also result in an increase in adoption of electric vehicles as well. Random forest is able to capture the general trend of population from counties from high to medium EV adoption. But lacks the predictive capability for counties whose EV population is low to none.

TABLE XI
LINEAR REGRESSION FEATURE IMPORTANCE

Feature	Coefficients
GDP	6017.124
Population	10201.080
Population Density	1516.667
Total Chargers	36330.774

Linear Regression's highest coefficient is total chargers, followed by population, GDP, and population density as shown in Table XI. All features are positive coefficients, indicating that any value or presence of these feature are associated with an increase of their respective coefficient values. With total chargers contributing the most to each prediction, with each charger adding 36,330.77 units of electric vehicles. Linear Regression continues the trend of total chargers being the most important feature, just like random forest and XGBoost models. When observing the actual values compared to predicted values of the Linear Regression model, Linear Regression is able to predict the trend of increasing EV populations across counties, however it is unable to stay accurate to the actual EV population. Linear Regression performs similarly to Random Forest as well, as it is unable to predict EV populations for counties that little to no electric vehicles.

TABLE XII
SVR PERMUTATION IMPORTANCE

Feature	Permutation Importance
GDP	-0.002969
Population	0.007604
Population Density	-0.002969
Total Chargers	-0.011504

Originally, predictions for SVR was conducted without a grid search, however it resulted in extremely poor performance

and a negative R^2 value of -0.01%, meaning that the predictions of the algorithm are completely random. However, using grid search, the better parameters were found for the model. With the new parameters of C - 20000, Epsilon - 1e-06, and a Linear Kernel, it has only resulted in a model that performs poorly, is below expectations, and worse than all other previously mentioned models.

As displayed on table XV, total chargers being the highest permutation, a consistent trend with all the models used for this research. In the SVR model, there are negative importance scores for the GDP and population density features. Indicating that these features detract from the performance of the model when included to use in its predictions. Table XVII displays the actual and predicted values in the form of a table.

V. CONCLUSION AND FUTURE WORK

From this research, it shows that XGBoost is the most effective model with the highest predictive capabilities and accuracy. Followed closely by Random Forest. Linear Regression, despite having a high variance, still has a low accuracy score, suggesting it is unsuitable for predicting EV population. This is also the case for SVR, despite using a grid search to find the best parameters, only an R^2 of 0.532 was obtained, not meeting the requirements for an effective predictive model. Additionally, the feature that was heavily relied upon by all four models were 'Total Chargers'. This is due to the fact that the total amount of chargers within a county is directly proportional to the demand of charging, which in this case, demand will always increase with a large population of EVs within the area. Population and population densities are not always a directly reliable feature to use, as for example, Los Angeles County has nearly the same density and population as San Francisco County, however due to different cultures and governmental policies, Los Angeles has a more widespread EV population.

Future work would explore the addition of polynomial features, hyper-parameter tuning, and the incorporation of more features closely related to EV infrastructure, as the 'Total Chargers' feature was primarily used in all four predictive models. Lastly, expanding the dataset to other regions would provide more data for the models to understand the different scaling and relationships between features as well.

SUPPLEMENTARY CODES

All the codes used in this paper can be accessed through the following Github and Google Colab links:

GitHub Repository, Washington, California, and Florida Datasets, Texas and New York Datasets, Models

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Oneal Waseem, Sir Tyrone Raphael Joseph Vargas, Xian Onnodhia Nanhai Habibie: Conceptualization, Data Curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Nunung Nurul Qomariyah, Raymond Bahana:** Conceptualization, Writing - Review &

Editing, Supervision, Project administration, Funding acquisition.

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